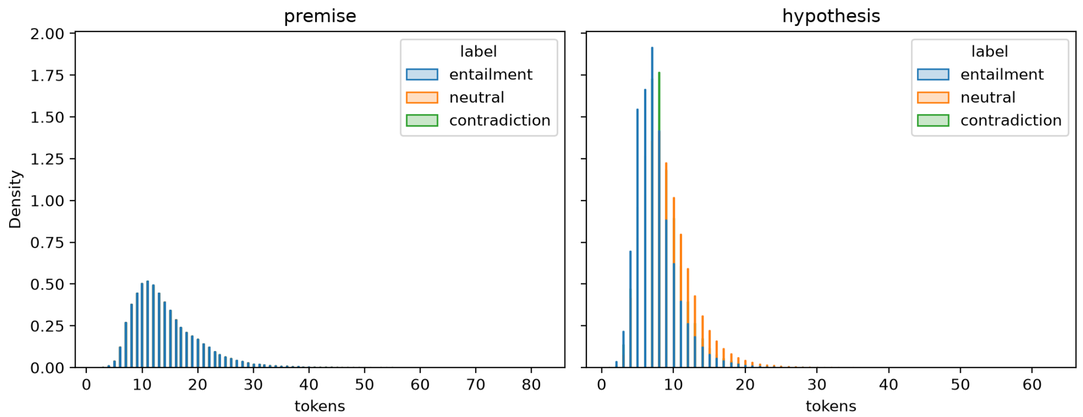
NLP1 Project 1 — NLI on SNLI

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github.com/neelshahx/nlp1-2026

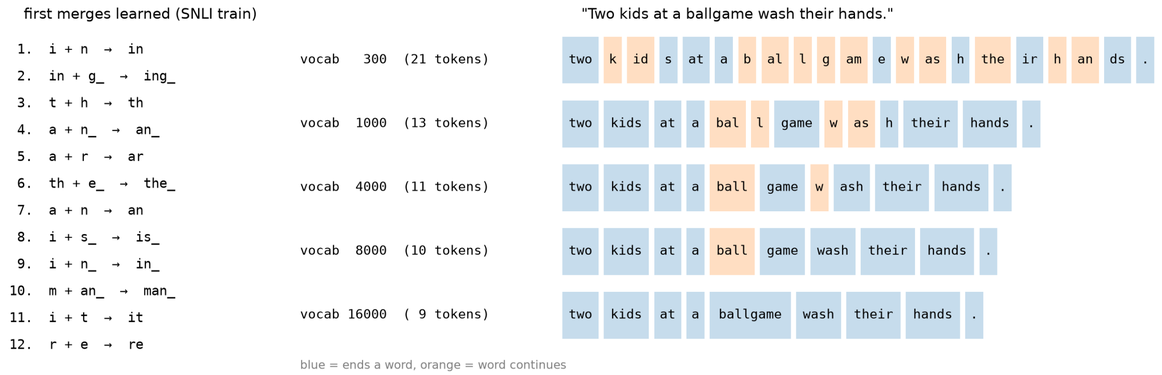
Q1 · Task & data

- (premise, hypothesis) → entailment / neutral / contradiction
- SNLI: 549k train / 9.8k dev / 9.8k test, balanced (~33% each)
- Artifacts (Gururangan et al. 2018): negation → contradiction, longer neutral hypotheses
- Token overlap (Jaccard): ent 0.31 · neu 0.26 · con 0.22
- PMI(token, label) = log p(token | label) / p(token)
- Setup: stratified 5-fold CV on train (seed 0); dev for selection; test once

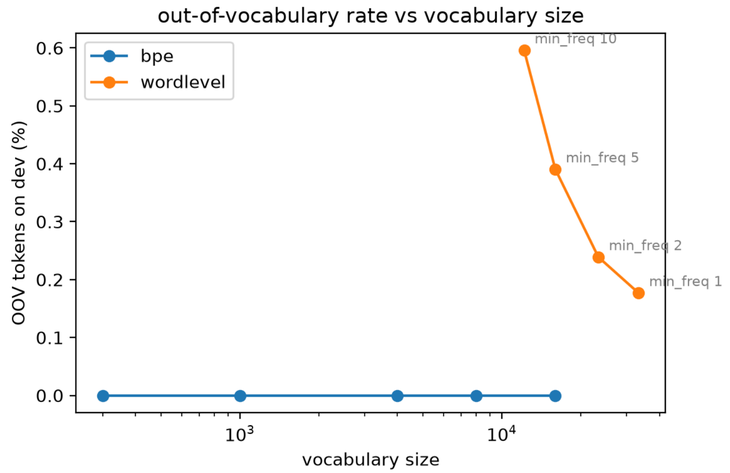


Q2 · Tokenizer

- BPE (HF tokenizers), min_freq 2, trained on SNLI train (premise + hypothesis) only
- 549k pairs is enough data: no external corpus needed
- Vocab 4k / 8k / 16k → CV acc 64.0 / 64.2 / 63.9: flat
- Word-level baseline 63.9 vs BPE 64.3 → inside ROPE; BPE wins on coverage, not accuracy
- Subwords → 0% out-of-vocabulary on dev (word-level 0.2–0.6%)
- Tokenizer is fit on all of train (incl. each held-out fold), but uses no labels



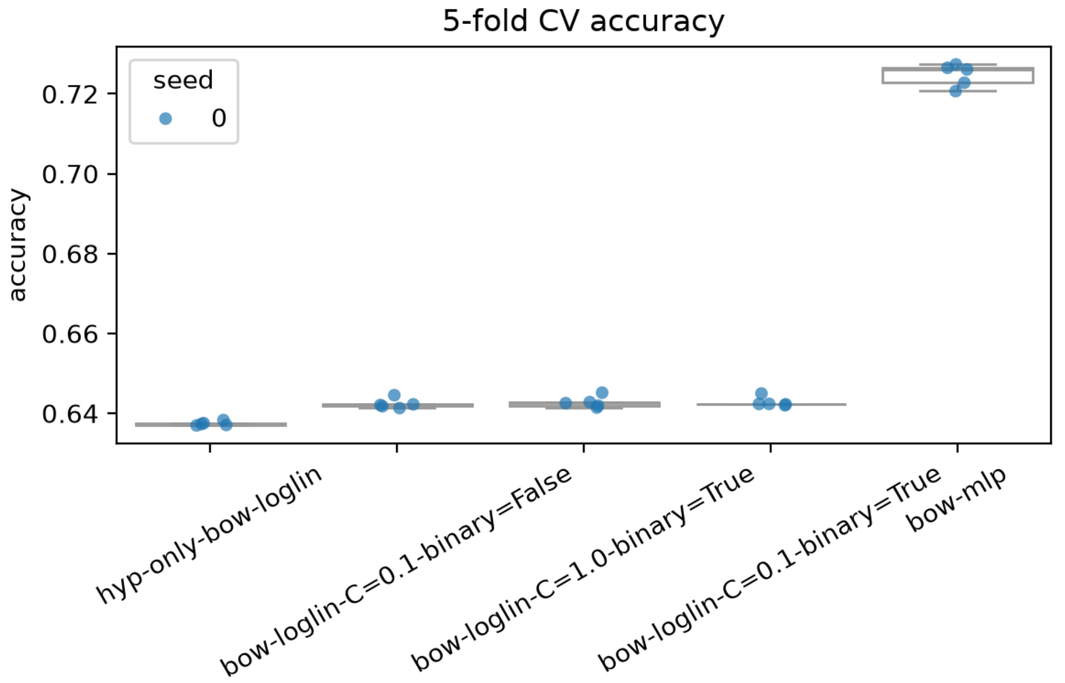
BPE merges the most frequent adjacent pair first (left). One dev hypothesis at growing vocabulary sizes (right): 21 pieces at 300, one token per word by 16k; the rare word is the last to merge.



Share of dev tokens the tokenizer maps to [UNK]. BPE falls back to characters so it never fails; word-level cannot.

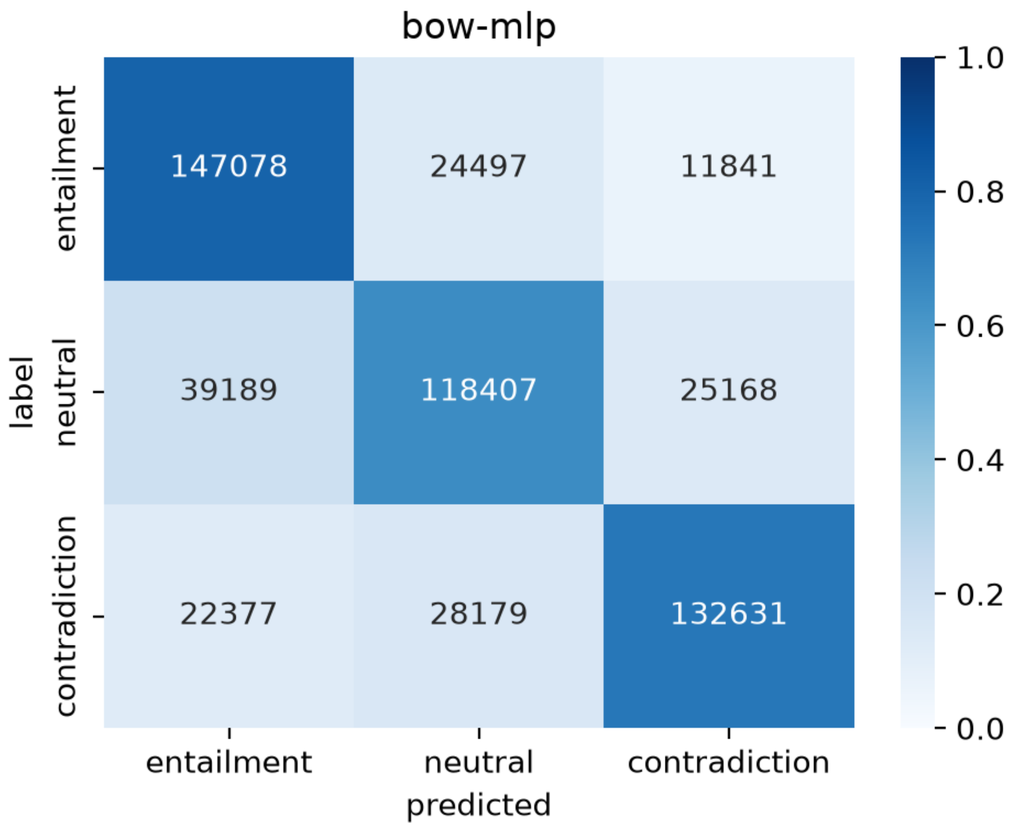
Q3 · Representation

- Both sides at once: concat [BoW(P) ; BoW(H)], so the model knows which side a token is on
- Joint (one bag over "P [SEP] H" = BoW(P) + BoW(H)) 58.8 vs concat 64.3: –5.5 pt, side information lost
- Hypothesis-only 63.7 vs concat 64.2, P(rope) = 0.997
- Indicators vs counts: 64.28 vs 64.24 → inside ROPE (short sentences)
- Linear on concat = f(P) + g(H): no P × H interaction
- n-grams 1–2 (local word order): unigrams 64.2, bigrams TBD → next: TF-IDF, cross features |u – v|, u ⊙ v



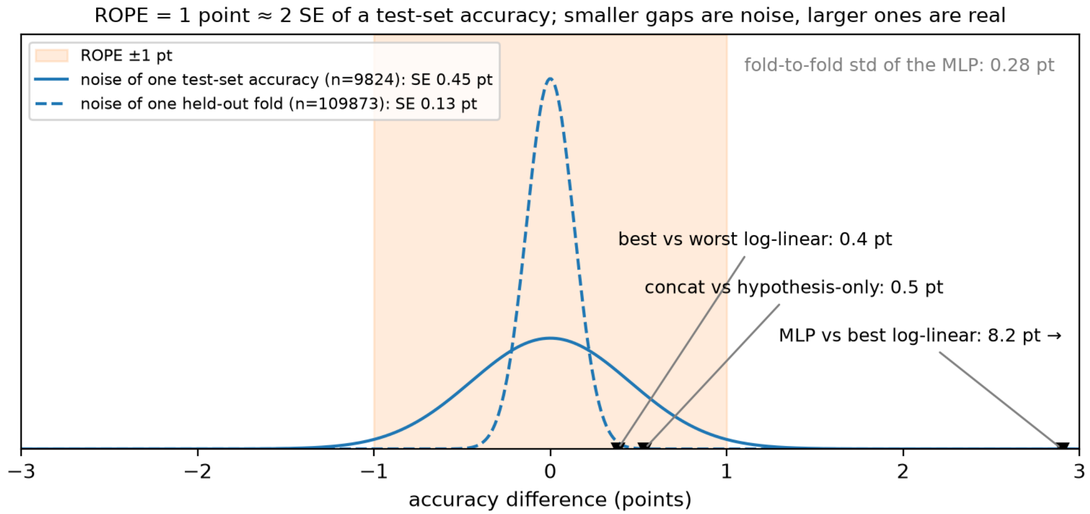
Q4 · Architectures

- Log-linear (lbfgs, L2), C ∈ {0.1, 1, 10}: flat ≈ 64%
- MLP: 16k → 128 → ReLU → dropout 0.1 → 3
- Adam 1e-3, weight decay 1e-4, early stopping (10% val, patience 3)
- MLP 72.5% (+8 pts): hidden layer mixes P and H
- Third: residual MLP (LayerNorm + skip), blocks {0, 2}, hidden {128, 512} — TBD

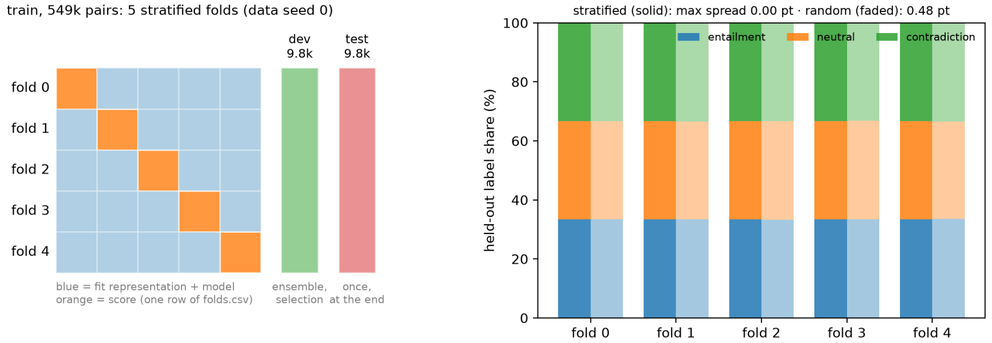


Q5 · Evaluation

- Stratified 5-fold CV, same folds for all → paired; equal label share per fold (random split: ±0.5 pt)
- Accuracy (+ macro-F1; balanced, so ≈ equal)
- Floors: majority 33%, hypothesis-only 63.7%
- Correlated Bayesian t-test (baycomp, ρ = 1/k)
- ROPE = 1 accuracy point ≈ 2 SE of a test-set accuracy (SE ≈ 0.45 pt)
- Neutral is the hardest class



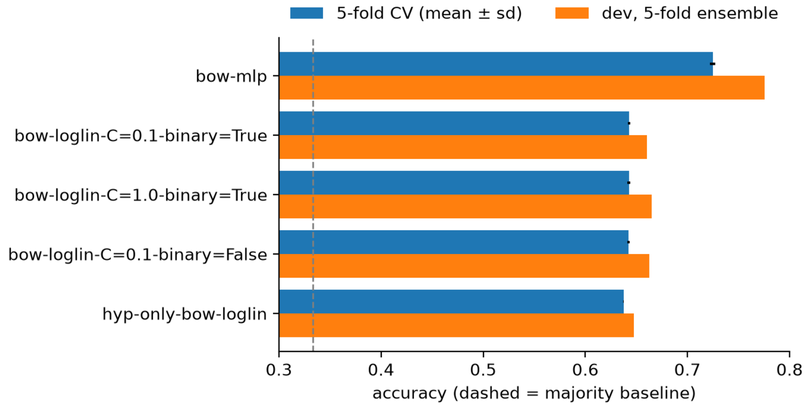
ROPE ±1 pt next to the sampling noise of one evaluation. The log-linear gaps sit inside it; the MLP gap is far outside.

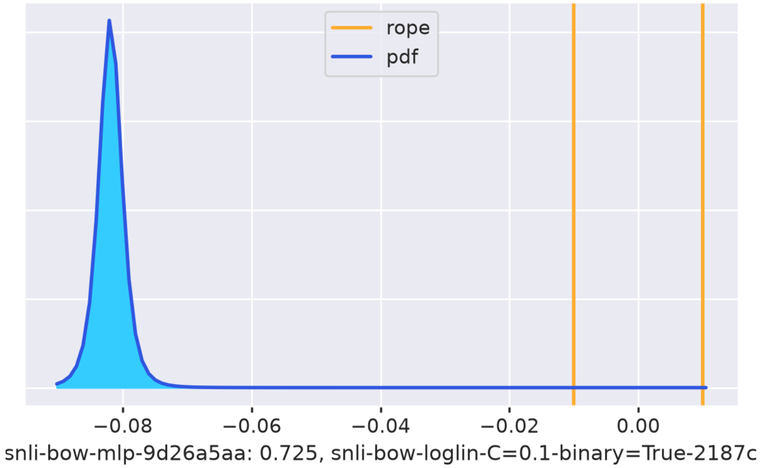


Every experiment scores the same 5 stratified folds (paired). Stratified folds keep label shares identical; a random split drifts by ±0.5 pt.

Q6 · Best model

- BoW + MLP: CV 72.5 ± 0.3
- P(MLP > log-linear) = 1.00, P(rope) = 0
- Log-linear variants all inside ROPE → pick simplest
- Dev, 5-fold ensemble: MLP 77.6 vs log-linear 66.5
- Test: once, final model only — TBD





Posterior of (log-linear – MLP) accuracy from the correlated t-test: all mass left of the ROPE.